Zachary Gameiro

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Technical Skills

Programming: Python, JavaScript, TypeScript, SQL, C/C++
Scientific Computing: PyTorch, clustering, graph embeddings, DBSCAN/HDBSCAN, data preprocessing, experimental analysis
Wet Lab Techniques: micropipetting, serial dilutions, buffer/solution prep, reagent weighing, pH adjustment, centrifugation, vortexing, sterile technique
Molecular / Biochemistry Methods: bacterial culture, cell culture, plating/streaking, colony counting, transformation/transfection, DNA extraction, gel electrophoresis, agarose gels, SDS-PAGE, Western blot, ELISA
Bioinformatics / Visualization: microscopy, fluorescence microscopy, PDB parsing, molecular structure visualization, microscopy data workflows, secondary-structure handling
Data & Documentation: lab notebook documentation, Excel/Python analysis, statistics/graphing, Git, Linux, Docker, structured extraction, validation
Safety: WHMIS, biosafety, protocol-driven lab work
Languages: English, French, Portuguese

Education

Simon Fraser University — Burnaby, BC
Bachelor of Science, Joint Major in Computing Science and Microbiology & Biochemistry — Expected May 2027

Course Laboratory Experience

Biochemistry, Organic Chemistry, and Genetics Labs — SFU

- Completed course-based wet-lab work involving **micropipetting**, serial dilutions, buffer preparation, centrifugation, vortexing, sterile technique, and protocol-driven sample handling.
- Performed molecular and biochemistry workflows including bacterial culture, plating/streaking, colony counting, transformation/transfection, **DNA extraction**, agarose gel electrophoresis, **SDS-PAGE**, **Western blot**, and **ELISA**.
- Documented experiments in lab notebooks and used Excel/Python plus statistics/graphing workflows to analyze and interpret results.

Scientific Computing Projects

PDB Structure Viewer — Personal Prototype — 2025

- Prototyped a browser-based Protein Data Bank viewer that renders helices and sheets while handling secondary-structure annotations.
- Explored frontend versus backend parsing approaches to improve responsiveness for interactive molecular visualization.

Receipt OCR — Personal Project — 2025
Python, PaddleOCR, Docker, Google Cloud Run, LLM | GitHub

- Built a **Python** pipeline that turns noisy visual inputs into structured records through OCR, layout reconstruction, and validation.
- Improved throughput by roughly **200%** versus the earlier prototype while keeping outputs consistent across varied document layouts.

Academic Projects

SMLM Clustering with Graph Embeddings — SFU School Team Project — 2025
Python, PyTorch, MIRO, DBSCAN, HDBSCAN | GitHub

- Co-built a single-molecule localization microscopy pipeline using MIRO graph embeddings and density-based clustering.
- Reproduced and evaluated clustering workflows on noisy biological point-cloud data to improve recovered structure quality.

Ferry Terminal Booker — School Team Project — 2025
C, File I/O, Disk Storage, CLI | GitHub

- Co-built a file-backed reservation system with validation, search flows, and reliable stored-record updates.
- Implemented safe record-handling patterns to keep user actions and persisted data consistent.

Experience

YMCA of Northern Alberta — Lifeguard Sep 2019 – Jun 2020

- Ensured swimmer safety, monitored facilities, and executed emergency response procedures in a regulated aquatic environment.
- Performed pool chemistry checks, including **pH testing** and related water-quality verification, as part of routine facility operations.

Fluor Corporation (LNG Canada) — General Labourer Sep 2020 – Jan 2024

- Completed safety-critical work across active industrial site phases while following strict procedural controls.
- Coordinated with crews under schedule pressure in environments where reliability and compliance mattered.